Lexical exceptionality in paradigm-specific learning:

Modeling stem-final obstruent alternations in Korean verbs and adjectives

Stella Eunsoo Hong

Department of Linguistics, University of Utah

SCiL 2026
July 3–4th, 2026

Contents

Question of Interest

- **Lexical Exceptionality**
  - Learners must identify exceptional forms and posit item-specific grammars departing from a general pattern.

- **Within Optimality Theory**
  - **Cophonology** (Inkelas and Zoll, 2003)
    - Separate rankings for different morphological classes
      Class X: Constraint A $\gg$ Constraint B
      Class Y: Constraint B $\gg$ Constraint A
  - **Lexical Indexation** (Pater, 2000)
    - Constraints indexed to lexical classes
      Constraint $A_{\text{class X}}$ $\gg$ Constraint B $\gg$ Constraint $A_{\text{class Y}}$

Question of Interest

- **Too Many Solutions Problem**
  - Exuberance in solving markedness violations (McCarthy, 2002)
  - Different solutions to $^{*}\mathrm{N}\mathring{\mathrm{C}}$ (Pater, 1999)
    - /mp/ $\rightarrow$ [mb]: violates $\textsc{Ident(voice)}$
    - /mp/ $\rightarrow$ [bp]: violates $\textsc{Ident(nasal)}$
    - /mp/ $\rightarrow$ [m]/[p]: violates $\textsc{Max-C}$

- **Lexical Exceptionality X Multiple Repair Strategies**
  - When multiple repair strategies coexist for lexically exceptional patterns, how do learners simultaneously identify exceptional items and learn the corresponding repair strategy?

Korean Stem-final Conjugation

- A useful test case for the interaction of lexical exceptionality and multiple repair strategies
- Three alternating classes undergo intervocalic lenition, each exhibiting a distinct repair strategy:
    - /p/ → [w]
    - /t/ → [ɾ]
    - /s/ → ∅
- Non-alternating classes remain faithful.

Class	Stem	Conjugation
/p/-alternating	/ka.ka**p**/ 'close'	[ka.ka.**w**-ʌ]
/p/-non-alternating	/i**p**/ 'wear'	[i.**p**-ʌ]
/t/-alternating	/kjɛ.da**t**/ 'realize'	[kjɛ.da.**ɾ**-a.jo]
/t/-non-alternating	/mu**t**/ 'bury'	[mu.**t**-ʌ]
/s/-alternating	/na**s**/ 'better'	[na-a.do]
/s/-non-alternating	/u**s**/ 'laugh'	[u.**s**-ʌ]

Korean Stem-final Conjugation

- Suffixes provide the phonological environment triggering alternation.
  - Before consonant-initial suffixes, the intervocalic environment is absent.
  - The contrast between alternating and non-alternating classes is no longer observable.
- *Lexical idiosyncrasy resides in the stem*

Class	Stem	Conjugation
/p/-alternating	/ka.ka**p**/ 'close'	[ka.ka**p**-ta]
/p/-non-alternating	/i**p**/ 'wear'	[i**p**-ta]
/t/-alternating	/kjɛ.da**t**/ 'realize'	[kjɛ.da**t**-ta]
/t/-non-alternating	/mu**t**/ 'bury'	[mu**t**-ta]
/s/-alternating	/na**s**/ 'better'	[na**s**-ta]
/s/-non-alternating	/u**s**/ 'laugh'	[u**s**-ta]

- Lee (2013) analyzes Korean stem-final obstruent alternations using lexical indexation.
- A shared markedness constraint:

$$*\text{V-O}{\small\text{BS}}\text{-V}$$

- Ranking paradox:

$$*\text{V-O}{\small\text{BS}}\text{-V} \gg \text{F}{\small\text{AITH}} \qquad \text{(alternating)}$$

$$\text{F}{\small\text{AITH}} \gg *\text{V-O}{\small\text{BS}}\text{-V} \qquad \text{(non-alternating)}$$

- Solution: constraint cloning (Pater, 2009)

$$*\text{V-O}{\small\text{BS}}\text{-V}_{\text{Alter}} \gg \text{F}{\small\text{AITH}} \gg *\text{V-O}{\small\text{BS}}\text{-V}_{\text{Non-Alter}}$$

- **Unaddressed:** How do paradigm-specific instantiations of the lexical-indexation schema combine into a single grammar?
- Different repair strategies violate different FAITH constraints:
  - /p/-alternating stems violate ID-C(ONSONANTAL)
  - /t/-alternating stems violate ID-LAT(ERAL)
  - /s/-alternating stems violate MAX(-C)
- **Expanded Ranking:** FAITH constraint implicated in a given repair must be outranked by those associated with the competing repair strategies.

$$*\text{V-Obs-V}_{P}^{Alt} \gg \text{ID-Lat},\ \text{Max} \gg \text{ID-C} \gg *\text{V-Obs-V}_{P}^{NonAlt}$$

$$*\text{V-Obs-V}_{T}^{Alt} \gg \text{ID-C},\ \text{Max} \gg \text{ID-Lat} \gg *\text{V-Obs-V}_{T}^{NonAlt}$$

$$*\text{V-Obs-V}_{S}^{Alt} \gg \text{ID-C},\ \text{ID-Lat} \gg \text{Max} \gg *\text{V-Obs-V}_{S}^{NonAlt}$$

Lexically-Scaled MaxEnt

A probabilistic implementation of constraint indexation (Linzen et al., 2013; Hughto et al., 2019)

$$\mathcal{H}_i = \sum_{\gamma \in \Gamma} \left(w_\gamma + \sum_{m \in \mu_i} s_{\gamma m} \right) v_{\gamma i}$$

($\underline{\Gamma}$: constraint set; $\underline{\mathcal{H}_i}$: candidate harmony; $\underline{w_\gamma}$: general constraint weight; $\underline{s_{\gamma m}}$: morpheme-specific scaling factor; $\underline{v_{\gamma i}}$: violations of γ by candidate i)

The additive penalty $s_{\gamma m}$ serves as a probabilistic analogue of lexical indexation.

1. Alternating classes receive increased scaling on *V-OBS-V, yielding *V-OBS-V $\gg$ FAITH. Non-alternating classes receive no comparable scaling, consistent with FAITH $\gg$ *V-OBS-V.

2. Repair-specific FAITH constraints receive the smallest scaling:
   - ID-C(ONSONANTAL) (/p/-alternation)
   - ID-LAT(ERAL) (/t/-alternation)
   - MAX(-C) (/s/-alternation)
3. Only idiosyncratic stems receive nonzero scaling; suffixes remain unscaled.

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	4	6	6	6
p-alternating	6	3	2	0
p-non-alternating	0	1	1	1
t-alternating	6	3	0	4
t-non-alternating	0	1	1	1
s-alternating	6	0	3	3
s-non-alternating	0	1	1	1
suffixes	0	0	0	0

Idealized learning outcome

- **Toy stems:** generated in a CV-/p,t,s/ structure
  - Reflect Korean phonotactics (no consonant clusters)
  - Segment inventories based on the NIKL (National Institute of Korean Language) Romanization system
    - C $\in$ {g, kk, k, d, tt, t, b, pp, p, j, jj, ch, s, ss, h, n, m, r}
    - V $\in$ {a, eo, o, u, eu, i, ae, e, oe, wi}
  - Stem class proportions mirror type frequencies reported in Lee (2013)

p-alternating	68 (72.3%)
p-non-alternating	5 (5.3%)
t-alternating	5 (5.3%)
t-non-alternating	7 (7.5%)
s-alternating	5 (5.3%)
s-non-alternating	4 (4.3%)

Stem Class Distribution in the Sejong Corpus

Complete inventory of toy stems

Class	Stems
p-alternating	neop- ppaep- sep- ssep- gop- bwip- seup- dep- dop- baep- nep- kkeup- moep- kkip- kkop- ssaep- chap- rep- kop- teup- cheop- tep- heup- rwip- jjwip- paep- bop- bup- mip- dwip- choep- sop- swip- pwip- jjeop- keop- soep- ssap- taep- mwip- rup- chwip- bep- peup- ppep- chop- keup- geup- nip- ppip- saep- noep- kkap- jjup- gwip- jjep- meup- ttop- jep- daep- joep- mop- mep- ssoep- ttaep- reop- jjop- ppap-
p-non-alternating	jaep- kkep- jjip- rap- toep-
t-alternating	tat- jat- ket- pot- chit-
t-non-alternating	ppeut- goet- jot- pput- nat- kkut- sit-
s-alternating	ges- kkaes- hoes- koes- nos-
s-non-alternating	poes- meos- jjaes- ches-

Experiment

- **Toy suffixes:** designed to control exposure to intervocalic lenition environments
  - Alternation-triggering (-V) and non-triggering (-CV) suffixes were equally represented
  - Initial inventory consisted of two suffixes: *-a* and *-ta*
  - Additional V-/CV pairs were incrementally introduced
  - Vowels were drawn from the same monophthong inventory used in stem generation

Condition	Suffix inventory
2-suffix	*-ta*, *-a*
4-suffix	*-ta*, *-a*, *-te*, *-e*
6-suffix	*-ta*, *-a*, *-te*, *-e*, *-tu*, *-u*
8-suffix	*-ta*, *-a*, *-te*, *-e*, *-tu*, *-u*, *-to*, *-o*
10-suffix	*-ta*, *-a*, *-te*, *-e*, *-tu*, *-u*, *-to*, *-o*, *-ti*, *-i*

Experiment

Balanced Condition

- The /p/-paradigm was excluded to create a balanced distribution of repair frequencies.
- This removes the dominant glide-formation pattern associated with /p/-alternation.
- Two repair strategies are available for resolving *V-OBS-V:
  1. Lateralization (violating ID-LAT)
  2. Deletion (violating MAX)
- Each UR is paired with three candidate SRs:

UR	SR	%	*V-OBS-V	MAX	ID-LAT
/tat-a$_{\text{T-Alter}}$/	(a) [tata]	0.0	1	0	0
	(b) [tara]	1.0	0	0	1
	(c) [taa]	0.0	0	1	0
/ges-ta$_{\text{S-Alter}}$/	(a) [gesta]	1.0	0	0	0
	(b) [gerta]	0.0	0	0	1
	(c) [geta]	0.0	0	1	0

Example simulation data in the balanced condition

Imbalanced Condition

- The /p/-paradigm is included, yielding a skewed distribution of repair frequencies.
- This introduces the dominant glide-formation pattern associated with /p/-alternation.
- In addition to lateralization and deletion, glide formation (violating ID-C) becomes available.
- Each UR is paired with four candidate SRs:

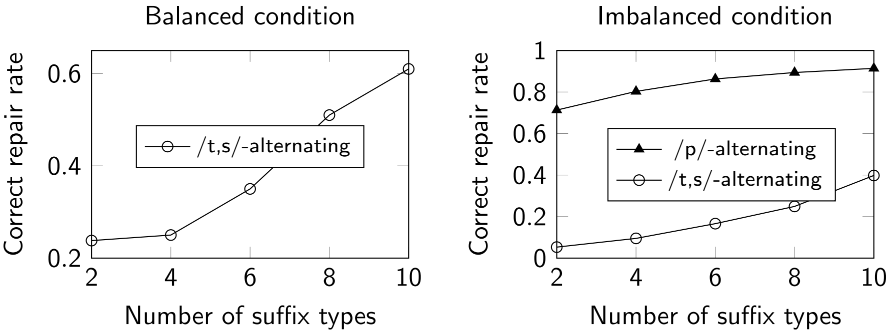
UR	SR	%	*V-OBS-V	MAX	ID-C	ID-LAT
/neop-a$_{\text{P-Alter}}$/	(a) [neopa]	0.0	1	0	0	0
	(b) [neowa]	1.0	0	0	1	0
	(c) [neola]	0.0	0	0	0	1
	(d) [neoa]	0.0	0	1	0	0
/tat-a$_{\text{T-Alter}}$/	(a) [tata]	0.0	1	0	0	0
	(b) [taja]	0.0	0	0	1	0
	(c) [tara]	1.0	0	0	0	1
	(d) [taa]	0.0	0	1	0	0

Example simulation data in the imbalanced condition

Results

Generalization improves with increasing suffix type frequency.

Evaluation focuses on repair accuracy of alternating stems in intervocalic contexts, i.e., when followed by -V suffixes.



→ This progression is reflected in the distribution of scaling weights.

Results

Stage 1: When given only few suffix types, scaling weights are misattributed to non-idiosyncratic suffixes.

	*V-Obs-V	Max	ID-Lat
General weights	0	0.79	0.79
t-alternating stems	0	0	0
t-non-alternating stems	0	0	0
s-alternating stems	0	0	0
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	2.16	2.16

Balanced condition (2 suffixes)

	*V-Obs-V	Max	ID-Lat	ID-C
General weights	0.69	2.6	2.6	0
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	0
t-alternating stems	0	0	0	0
t-non-alternating stems	0	0	0	0
s-alternating stems	0	0	0	0
s-non-alternating stems	0	0	0	0
-V suffix	0.69	0	0	0
-CV suffix	0	1.92	1.92	4.51

Imbalanced condition (2 suffixes)

- In the 2-suffix condition, each stem is observed only twice, whereas the same suffixes occur across all stem classes.
- Scaling weights are consequently assigned to suffixes rather than stems.

Results

Stage 2: Scaling weights shift from suffixes to stems, enabling the learner to distinguish alternating and non-alternating classes.

	*V-Obs-V	Max	Id-Lat
General weights	2.2	3.26	3.26
t-alternating stems	0.37	0	0
t-non-alternating stems	0	0	0
s-alternating stems	0.37	0	0
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	0	0

Balanced condition (4 suffixes)

	*V-Obs-V	Max	Id-Lat	Id-C
General Weights	4.98	6.29	6.29	2.71
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	2.54
t-alternating stems	0.61	0	0	2.88
t-non-alternating stems	0	0	0	2.54
s-alternating stems	0.61	0	0	2.88
s-non-alternating stems	0	0	0	2.54
-V suffix	0	0	0	0
-CV suffix	0	0	0	1.53

Imbalanced condition (6 suffixes)

- As suffix diversity increases, stems occur in a wider range of contexts, making them more likely to be lexicalized.
- The learner first distinguishes alternating from non-alternating classes.
- Exception: /p/-alternating stems are not lexicalized.

Results

Stage 3: Paradigm-specific patterns are distinguished among alternating classes

	*V-Obs-V	Max	ID-Lat
General weights	1.72	4.04	4.04
t-alternating stems	3.43	1.14	0
t-non-alternating stems	0	0	0
s-alternating stems	3.43	0	1.14
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	0	0

Balanced condition (10 suffixes)

	*V-Obs-V	Max	ID-Lat	ID-C
General Weights	7.11	8.98	8.98	4.46
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	3.76
t-alternating stems	2.55	0.69	0	5.2
t-non-alternating stems	0	0	0	3.76
s-alternating stems	2.55	0	0.69	5.2
s-non-alternating stems	0	0	0	3.76
-V suffix	0	0	0	0
-CV suffix	0	0	0	0

Imbalanced condition (10 suffixes)

- Alternating classes are further differentiated according to their repair strategy.
- /t/- and /s/-alternating classes establish distinct rankings between ID-Lat and Max.
- Exception: /p/-alternating stems remain unlexicalized.

Results

Exception: Statistical dominance prevents lexicalization of /p/-alternating stems.

Stem class	Type freq.
p-alternating	**68 (72.3%)**
p-non-alternating	5 (5.3%)
t-alternating	5 (5.3%)
t-non-alternating	7 (7.5%)
s-alternating	5 (5.3%)
s-non-alternating	4 (4.3%)

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	7.11	8.98	8.98	4.46
p-alternating stems	**0**	**0**	**0**	0
p-non-alternating stems	0	0	0	**3.76**
t-alternating stems	2.55	0.69	0	**5.2**
t-non-alternating stems	0	0	0	**3.76**
s-alternating stems	2.55	0	0.69	**5.2**
s-non-alternating stems	0	0	0	**3.76**
-V suffix	0	0	0	0
-CV suffix	0	0	0	0

- The /p/-alternating class overwhelmingly dominates the stem distribution (72.3%).
- /p/ $\rightarrow$ [w] alternation is treated as a default repair strategy rather than a lexically idiosyncratic pattern.
  - The /p/-alternating stems remain unscaled.
  - ID-C receives high scaling weights in the remaining classes.

Discussion

Sensitivity to lexical statistics

- The model adopts a *least-cost* strategy in tracking lexical exceptionality.
- Statistically dominant morpheme classes are learned as the general pattern.
- Exceptionality is not always localized to the true exceptional class:
  - /p/-irregular stems are not treated as idiosyncratic.
  - Under limited morphological evidence, scaling weights are assigned to suffixes.

$\rightarrow$ How are suffix frequencies distributed in natural language?

Discussion

Absolute suffix type frequency seems unlikely to be the bottleneck

- The Korean Google Treebank contains 29 C-initial and 33 V-initial suffix types, far exceeding the largest suffix inventories considered in the simulation.

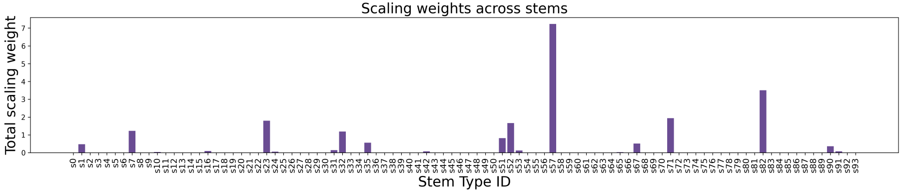
Context	Suffix inventory
-CV	-geona, -ge, -get, -go, -goyo, -guyo, -gi, -gillae, -na, -nayo, -neyo, -neureago, -neun, -neunda, -neundago, -neunde, -da, -daga, -dago, -daneun, -deogunyo, -deon, -dorok, -seumnida, -ja, -janeun, -jamaja, -ji, -jiman
-V	-n, -nde, -l, -m, -myeo, -myeon, -myeonse, -a, -ado, -aseo, -aya, -ayo, -at, -eo, -eodo, -eoseo, -eoya, -eoyo, -eot, -eotseot, -euna, -euni, -eureo, -eureogo, -eureoneun, -eumyeo, -eumyeon, -eumyeonse, -euseyo, -eusi, -eun, -eul, -eum

Suffix inventories in the Google Treebank Corpus (romanized)

Discussion

What seems to matter is paradigm saturation (Kodner, 2022)

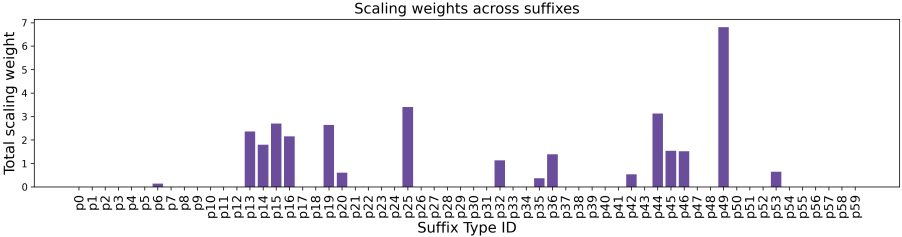
- Naturalistic input is highly Zipfian: few stems occur across many suffixes, while most occur with only a few.
- Only a small subset of stems received nonzero scaling weights, and even fewer acquired substantial scaling.



Discussion

What seems to matter is paradigm saturation (Kodner, 2022)

- In contrast, scaling factors were more widely assigned to suffixes, reflecting their broader attestation across stems.



Conclusion

To fully account for human language acquisition, two challenges remain:

- How can statistically non-dominant repair strategies be learned in the presence of overwhelmingly dominant patterns?
- How can the model correctly lexicalize the locus of exceptionality under conditions of limited paradigm saturation?

References

Hughto, C., Lamont, A., Prickett, B., and Jarosz, G. (2019). Learning exceptionality and variation with lexically scaled maxent. In Proceedings of the Society for Computation in Linguistics (SCIL) 2019, pages 91–101.

Inkelas, S. and Zoll, C. (2003). Is grammar dependence real. Ms, UC Berkeley and MIT. ROA-587.

Kodner, J. (2022). Computational models of morphological learning. In Oxford Research Encyclopedia of Linguistics.

Lee, P. (2013). Constraint cloning and lexical listing in korean irregular verbs and adjectives. Studies in Linguistics.

Linzen, T., Kasyanenko, S., and Gouskova, M. (2013). Lexical and phonological variation in russian prepositions. Phonology, 30(3):453–515.

McCarthy, J. J. (2002). A thematic guide to Optimality Theory. Cambridge University Press.

Pater, J. (1999). Austronesian nasal substitution and other nc effects. The prosody-morphology interface, pages 310–343.

Pater, J. (2000). Non-uniformity in english secondary stress: the role of ranked and lexically specific constraints. Phonology, 17(2):237–274.

Pater, J. (2009). Morpheme-specific phonology: Constraint indexation and inconsistency resolution.

Thank you for listening! ☻

Questions or comments?

stella.hong@utah.edu